Finite-Budget Structural Identifiability under Bounded Observation

Fabrizio De Palma
Independent Researcher

Abstract

We study the problem of structural regime identification under finite observational scale and finite computational budget.

Consider a stochastic or deterministic process observed through a representation map O_m of scale m, generating an observable sequence $Y_t \in \{0,1\}^m$. We define the novelty curve $N_m(t)$ as the number of distinct observations seen up to time t, and show that its finite-sample behavior encodes structural information about the underlying regime.

Our main result establishes a non-asymptotic equivalence between finite-budget identifiability and observational separability. Specifically, we derive:

- a sufficient condition for regime identification expressed through a critical time

$$t^*(m;\delta) = t_0(m) + C\,\frac{w(m)}{g(m)^2}\log\frac{1}{\delta},$$

where $t_0(m)$ captures structural latency, $w(m)$ effective integration cost, and $g(m)$ the maximal novelty gap between regimes;

- a matching lower bound showing that without positive separability, no test can outperform random guessing;
- the existence of a maximal identifiable observational scale under finite budget;
- and a structural allocation principle determining the optimal resolution compatible with identifiability.

The analysis reveals a phenomenon of resolution-induced collapse: increasing observational granularity may destroy identifiability under fixed resources. Identification limits thus arise not only from computational complexity but from structural properties of observable novelty growth.

We provide explicit regime constructions illustrating latency-dominated, estimation-dominated, and separation-dominated domains, and supply an artifact that operationalizes empirical estimation of the structural quantities.

Our results offer a solver-agnostic framework for analyzing finite-budget inference as a problem of structural observability rather than algorithmic power.

Keywords: structural identifiability, finite-budget inference, bounded observation, novelty growth, non-asymptotic analysis, hypothesis testing, observational scale, identifiability limits, statistical separation, structural collapse

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1. Introduction

Finite computational resources constrain not only how long we can compute, but how much structure we can reliably identify. In many settings—search, exploration, learning, adaptive control—the observable process is not the full state but a projection of it. Under bounded budgets, identification of structural properties becomes a question of observational geometry rather than raw computational power.

This paper studies the following problem:

Given a process observed through a finite-resolution map O_m and a finite budget b, when is structural regime identification possible?

We approach this question by analyzing the behavior of the observable novelty curve

$$N_m(t) = |\{O_m(X_1), \dots, O_m(X_t)\}|,$$

which counts the number of distinct observed configurations up to time t. The growth of $N_m(t)$ captures how structural information becomes visible under scale m.

Our central thesis is that finite-budget identifiability is governed by three structural quantities:

- **Latency** $t_0(m)$: the time before structural behavior stabilizes;
- **Integration window** $w(m)$: the effective correlation length governing novelty increments;
- **Separation gap** $g(m)$: the maximal expected difference in novelty growth between candidate regimes.

These quantities determine a critical identification time

$$t^*(m;\delta) = t_0(m) + C\frac{w(m)}{g(m)^2}\log\frac{1}{\delta},$$

yielding a non-asymptotic characterization of when regime identification is achievable under finite budget.

1.1 From Complexity to Observability

Traditional limitations in computer science are typically framed in terms of computational complexity: time, space, hardness reductions, or oracle separations. While such analyses address worst-case computational effort, they often abstract away the observational interface between the algorithm and the underlying process.

In contrast, we focus on a different type of barrier:

Structural identifiability under bounded observation.

Even when the underlying process is computationally simple, structural differences may remain indistinguishable if:

- the observational scale is too fine or too coarse,
- distinguishing events are too rare,
- novelty saturates too early,
- or integration costs exceed available budget.

Thus identifiability depends jointly on scale and budget.

This perspective reframes finite-budget search and inference as problems of structural observability rather than purely algorithmic design.

1.2 Observational Scale as a Structural Parameter

Let O_m denote a representation map of scale m. Increasing m:

- enriches representational capacity,
- but may increase latency,
- enlarge effective window size,
- and dilute separation between regimes.

We show that under mild monotonicity conditions there exists a maximal identifiable scale

$$m_c(b, \delta),$$

beyond which regime identification becomes impossible under budget b.

This phenomenon—resolution-induced collapse—implies that more detailed observation does not necessarily improve inference under resource constraints.

1.3 Main Contributions

The paper establishes the following results.

1. **Non-asymptotic sufficiency bound.**
   Under a block concentration assumption, regime identification with error at most δ is possible whenever $b \geq t^*(m; \delta)$.
2. **Matching lower bound.**
   If observational separability vanishes, no test can outperform random guessing, regardless of computational strategy.
3. **Maximal identifiable scale.**
   Under finite budget and confidence level, there exists a largest observational scale compatible with identification.
4. **Structural allocation principle.**
   The optimal observational resolution under budget constraint is determined by structural quantities, not algorithmic tuning.
5. **Explicit regime constructions.**
   We provide concrete examples illustrating latency-dominated, estimation-dominated, and separation-dominated domains.
6. **Operational artifact.**
   We supply a reference implementation that estimates structural quantities empirically and visualizes finite-budget identifiability.

1.4 Relation to Existing Work

The analysis connects to several established themes:

- finite-sample hypothesis testing,
- sample complexity in learning theory,
- bias–variance trade-offs,
- information-theoretic limits of inference,
- novelty growth in exploration processes.

However, our formulation differs in three respects:

1. It operates directly on novelty growth curves rather than likelihood ratios or model classes.
2. It makes observational scale explicit as a structural parameter.
3. It derives identifiability bounds from geometric properties of observable partitions.

The results are solver-agnostic and apply independently of specific algorithmic strategies.

1.5 Scope and Positioning

This work does not propose a new search algorithm.
It does not analyze worst-case computational hardness.
It does not assume specific stochastic models.

Instead, it isolates structural observability as a fundamental constraint on finite-budget inference.

The focus is two-regime identification under fixed representation. Extensions to multi-regime settings and adaptive scale refinement are left to future work.

1.6 Paper Organization

The remainder of the paper is structured as follows:

- Section 2 introduces the formal setup and observational framework.
- Section 3 defines structural regimes.
- Section 4 formalizes the regime testing problem.
- Section 5 derives a non-asymptotic upper bound for identification.
- Section 6 establishes a matching lower bound.
- Section 7 analyzes the critical time and its dependence on scale.
- Section 8 derives the maximal identifiable scale and optimal allocation principle.
- Appendices provide full proofs, structural examples, and artifact specification.

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2. Regimes under Bounded Observation

We consider a discrete-time stochastic process $(X_t)_{t\geq 1}$ evolving on a (possibly large or unknown) state space. The process is not directly observed. Instead, we access it through a finite-resolution representation

$$O_m: \mathcal{X} \to \{0,1\}^m,$$

where m denotes the observational scale.

Finite-budget testing under degraded observation has been studied in communication- and quantization-constrained settings (e.g., distributed hypothesis testing under rate constraints). In those models, fundamental limits arise from restricted information transmission **(Szabó et al., 2023; Inan et al., 2022)**. Here, the restriction is structural rather than communicational: the representation scale itself acts as the informational bottleneck.

Let

$$Y_t := O_m(X_t)$$

be the observed process, and define the bucket growth curve

$$N_m(t) := |\{Y_1, \dots, Y_t\}|,$$

i.e., the number of distinct observed patterns encountered up to time t.

The function $N_m(t)$ captures how rapidly new observational classes appear under scale m. Its qualitative behavior reflects structural properties of the underlying process as filtered through O_m.

The novelty curve is closely related to support-size and unseen-species estimation problems studied in i.i.d. models **(Hao et al., 2020; Chakraborty et al., 2023)**. Unlike those works, which aim at estimating support cardinality or missing mass, we use novelty growth as a regime-discriminative structural signature under representation constraints.

We now formalize two canonical regimes.

2.1 Stabilizing Regime

We say that the process is in a **stabilizing regime at scale** m if there exists a finite time $t_0(m)$ and a finite bound $K(m)$ such that

$$\mathbb{E}[N_m(t)] \leq K(m) \text{ for all } t \geq t_0(m).$$

Equivalently, the expected rate of novelty vanishes after $t_0(m)$:

$$\mathbb{E}[N_m(t+1) - N_m(t)] \to 0 \text{ as } t \to \infty.$$

In this regime, new observational patterns cease to appear after a finite horizon. The observed dynamics effectively collapse onto a finite subset of buckets.

Typical structural causes include:

- latent absorbing regions,
- strong internal mixing within a finite equivalence class,
- symmetry-induced indistinguishability,
- projection-induced collapse under coarse observation.

Importantly, stabilization is defined purely in terms of the observed curve $N_m(t)$, not in terms of hidden state properties.

2.2 Expanding Regime

We say that the process is in an **expanding regime at scale** m if there exists a function $h_m(t)$, strictly increasing in t, such that

$$\mathbb{E}[N_m(t)] \geq h_m(t) \text{ for all } t \leq b,$$

for some nontrivial time horizon b.

In particular, the expected novelty rate remains bounded away from zero over a finite window:

$$\exists \epsilon(m) > 0 \text{ such that } \mathbb{E}[N_m(t+1) - N_m(t)] \geq \epsilon(m)$$

for all $t \leq b$.

In this regime, new observational patterns continue to emerge under the given scale m. The structure does not collapse into a finite observable class within the relevant budget.

Structural mechanisms producing expansion include:

- persistent mixing across distinct equivalence classes,
- growth of reachable state regions,
- delayed manifestation of structural regularities,
- observation-sensitive differentiation.

2.3 Structural Interpretation

The distinction between stabilizing and expanding regimes is not algorithm-dependent. It arises from the interaction between:

- the intrinsic dynamics of X_t,
- the observational projection O_m,
- and the time horizon under consideration.

Crucially, a single underlying process may exhibit different regimes under different scales m. Coarser representations may induce early stabilization, while finer representations may preserve expansion over longer horizons.

Thus, regime classification is inherently **scale-relative**.

2.4 Remarks on Observational Relativity

The same hidden dynamics may appear stabilizing under one representation and expanding under another. This reflects a fundamental constraint: regime behavior is a property of the pair (X_t, O_m), not of X_t alone.

In subsequent sections, we analyze when two regimes are distinguishable under finite budgets and how this depends on structural separation induced by O_m.

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3. Structural Signatures under Finite Observation

The qualitative regime definitions of Section 2 can be refined into quantitative descriptors. These descriptors summarize how quickly structure manifests under observational scale m, how stable it is, and how well regimes separate.

We introduce three structural quantities associated with the observed curve $N_m(t)$.

3.1 Stabilization Time

For processes in the stabilizing regime, define the **stabilization time**

$$t_0(m) := \inf\{t \geq 1 : \mathbb{E}[N_m(s)] \leq K(m) \text{ for all } s \geq t\},$$

where $K(m)$ is the eventual saturation level (see Section 2.1).

Operationally, $t_0(m)$ measures how long it takes before the observable novelty rate effectively vanishes at scale m.

Even in expanding regimes, we may define $t_0(m)$ as the earliest time after which the novelty rate stabilizes below a chosen tolerance threshold.

Thus, $t_0(m)$ captures the **latency of structural revelation** under representation O_m.

3.2 Integration Window

Define the **integration window**

$$w(m)$$

as the minimal time interval required for reliable estimation of the local novelty rate.

Formally, let

$$\Delta_m(t) := N_m(t+1) - N_m(t),$$

and define $w(m)$ as the smallest window size such that empirical averages of $\Delta_m(t)$ over windows of length $w(m)$ concentrate around their expectation with prescribed accuracy.

Intuitively:

- Larger $w(m)$ means more temporal integration is required.

- Small $w(m)$ indicates rapid statistical stabilization.

The integration window quantifies **estimation complexity** at scale m.

The integration window w(m) plays a role analogous to mixing-time-dependent windowing in Markovian missing-mass estimation, where block aggregation is required for concentration under dependence **(Pananjady et al., 2024)**. Here, w(m) abstracts that structural necessity without assuming a specific process class.

3.3 Structural Separation Gap

Let R_0 and R_1 denote two regimes as defined in Section 2.

Define the **structural separation gap**

$$g(m) := \sup_{t \leq b} \mid \mathbb{E}_{R_1}[N_m(t)] - \mathbb{E}_{R_0}[N_m(t)] \mid.$$

The quantity $g(m)$ measures the maximal observable discrepancy between regimes at scale m within budget b.

When $g(m) > 0$, the regimes are structurally separable under O_m.
When $g(m) = 0$, they are observationally indistinguishable at that scale.

This gap is independent of any specific testing procedure. It is entirely determined by:

- the dynamics of the process,
- the observational projection O_m,
- and the time horizon.

3.4 Interplay Between the Signatures

The three quantities serve distinct roles:

- $t_0(m)$: latency before structural behavior stabilizes;
- $w(m)$: estimation horizon needed for reliable inference;
- $g(m)$: magnitude of regime separation.

In general:

- finer scales m may increase $t_0(m)$,
- may increase $w(m)$,
- and may decrease $g(m)$ if separability dilutes across features.

These interactions determine whether regime identification is feasible under a given finite budget.

3.5 Scale Relativity

The structural signatures depend explicitly on m.
A regime may be separable at coarse scale but indistinguishable at finer resolution, or vice versa.

Therefore, identifiability is not intrinsic to the underlying dynamics alone; it is jointly determined by:

$$(\text{dynamics}, O_m, b).$$

The next section formalizes the testing problem under finite budgets using these structural quantities.

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4. Testing under Finite Budget

We now formalize the regime identification problem under finite observational and temporal constraints.

4.1 Problem Formulation

Fix:

- an observational scale m,
- a finite budget $b \in \mathbb{N}$,
- two structural regimes R_0 and R_1,
- and confidence level $1 - \delta$, with $\delta \in (0,1)$.

We observe the prefix

$$Y_{\leq b} := (Y_1, \dots, Y_b), Y_t = O_m(X_t).$$

A (possibly randomized) test is any measurable function

$$\mathsf{T}_m : \{0,1\}^{m \times b} \to \{0,1\},$$

where output 0 corresponds to regime R_0 and output 1 to regime R_1.

The error probability of T_m is defined as

$$\mathcal{E}(\mathsf{T}_m) := \max \left\{ \Pr_{R_0}(\mathsf{T}_m = 1), \Pr_{R_1}(\mathsf{T}_m = 0) \right\}.$$

The testing problem is:

Determine whether there exists a test T_m such that
$\mathcal{E}(\mathsf{T}_m) \leq \delta$.

This problem depends only on:

- the induced observed laws under R_0 and R_1,
- the observational scale m,
- and the budget b.

It does not depend on any specific solver or exploration strategy.

4.2 Observational Separability

We say that regimes R_0 and $R_1$are **separable at scale m under budget b** if the induced distributions of the observed prefix $Y_{\leq b}$ differ.

Equivalently, separability requires

$$\exists t \leq b \text{ such that } \mathbb{E}_{R_0}[N_m(t)] \neq \mathbb{E}_{R_1}[N_m(t)].$$

When the structural separation gap

$$g(m) = \sup_{t \leq b} \mid \mathbb{E}_{R_1}[N_m(t)] - \mathbb{E}_{R_0}[N_m(t)] \mid$$

is strictly positive, separability is possible in principle.

When $g(m) = 0$, the regimes induce identical bucket-growth behavior and are observationally indistinguishable at that scale.

4.3 Finite-Budget Identifiability

We define **finite-budget identifiability at scale** m as the existence of a test T_m with error at most δ.

Formally:

$$\text{Scale } m \text{ is identifiable under } (b, \delta) \Leftrightarrow \exists \mathsf{T}_m \text{ such that } \mathcal{E}(\mathsf{T}_m) \leq \delta.$$

The key question of this paper is:

Under what structural conditions on
$t_0(m)$, $w(m)$, and $g(m)$
is identifiability achievable?

4.4 Budget–Scale Interaction

Two competing effects determine identifiability:

1. **Latency:**
   If $b < t_0(m)$, structural differences have not yet manifested.
2. **Estimation noise:**
   If b is insufficient relative to the integration window $w(m)$, empirical novelty estimates are unreliable.
3. **Separation magnitude:**
   If $g(m)$ is small, distinguishing regimes requires longer observation.

Thus identifiability is governed by a balance between:

$$\text{latency} + \text{estimation cost} + \text{separation strength}.$$

The next sections make this balance precise via non-asymptotic upper and lower bounds.

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5. Upper Bound: Finite-Budget Identifiability

We now establish a sufficient condition for regime identification under finite budget.

The result is non-asymptotic and explicitly characterizes the interplay between:

- stabilization latency $t_0(m)$,
- integration window $w(m)$,
- structural separation gap $g(m)$,
- confidence level δ.

5.1 Assumptions

Fix observational scale m.

Assume:

1. **Post-latency stability.**
   For all $t \geq t_0(m)$, the novelty increments

$$\Delta_m(t) := N_m(t+1) - N_m(t)$$

have bounded variance:

$$\mathrm{Var}\,(\Delta_m(t)) \leq \sigma_m^2.$$

2. **Window concentration.**
   Empirical averages over windows of length $w(m)$ satisfy standard concentration bounds (e.g., Hoeffding-type or Bernstein-type inequalities).
3. **Structural gap.**
   There exists

$$g(m) > 0$$

such that for some $t \leq b$,

$$| \mathbb{E}_{R_1}[N_m(t)] - \mathbb{E}_{R_0}[N_m(t)] | \geq g(m).$$

These assumptions depend only on the induced observed process and not on any specific solver.

5.2 Critical Identification Time

Define the **critical identification time**

$$t^*(m;\delta) := t_0(m) + C\frac{w(m)}{g(m)^2}\log\frac{1}{\delta},$$

where $C > 0$ is a universal constant depending only on the chosen concentration inequality.

Interpretation:

- $t_0(m)$: time until structural behavior stabilizes,
- $w(m)$: estimation window size,
- $g(m)$: magnitude of separability,
- $\log(1/\delta)$: confidence penalty.

As in classical testing theory, the quadratic dependence on the separation gap g(m) mirrors classical finite-sample testing rates, where discrimination cost scales as $1/g^2$ **(see, e.g., Tsybakov, 2009; Wainwright, 2019)**.

5.3 Main Sufficiency Result

Theorem 5.1 (Finite-Budget Sufficiency)

Fix scale m and confidence level $1 - \delta$.

If the budget satisfies

$$b \geq t^*(m;\delta),$$

then there exists a test T_m such that

$$\mathcal{E}(\mathsf{T}_m) \leq \delta.$$

Proof Sketch

For $t \geq t_0(m)$, consider the empirical estimate

$$\widehat{N}_m(t)$$

computed over windows of size $w(m)$.

By the concentration assumption,

$$\Pr\left(\mid \widehat{N}_m(t) - \mathbb{E}[N_m(t)] \mid \geq \frac{g(m)}{2} \right) \leq \delta$$

provided

$$t - t_0(m) \geq C\,\frac{w(m)}{g(m)^2} \log \frac{1}{\delta}.$$

Thus, once $b \geq t^*(m;\delta)$, the empirical gap between regimes exceeds half the structural gap with probability at least $1-\delta$, allowing correct classification by thresholding.

□

5.4 Interpretation

The bound reveals three structural barriers to identification:

- **Latency barrier:**
  If $b < t_0(m)$, structural differences have not yet manifested.
- **Estimation barrier:**
  If $b - t_0(m)$ is insufficient relative to $w(m)$, noise dominates.
- **Separation barrier:**
  Small $g(m)$ requires quadratic growth in observation length.

Finite-budget identifiability is therefore governed by the joint behavior of

$$(t_0(m), w(m), g(m)).$$

5.5 Structural Implications

The upper bound is:

- solver-independent,
- purely observational,
- non-asymptotic,
- scale-relative.

It shows that regime identification is feasible precisely when structural separation exceeds the combined latency and estimation costs induced by finite observation.

The next section establishes that this condition is also necessary in the absence of separability.

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6. Lower Bound: Structural Indistinguishability

The sufficiency result of Section 5 shows that identification is possible when the structural gap $g(m)$ is positive and the budget exceeds the critical time $t^*(m;\delta)$.

We now show that separability is also necessary:
if the structural gap vanishes under scale m, then no finite-budget procedure can distinguish the regimes.

The lower bound aligns with classical indistinguishability arguments: when induced observed laws coincide, no test can outperform chance. This parallels standard minimax lower bounds derived via total variation or KL arguments **(see, e.g., Wainwright, 2019)**.

6.1 An Explicit Indistinguishability Construction

Fix an observational scale $m \geq 1$.
Let $K := 2^m$.

Define a hidden state space

$$S = \{0,1\} \times \{0,1,\dots,K-1\}.$$

The first coordinate $i \in \{0,1\}$ encodes the regime label.
The second coordinate $j \in \{0,\dots,K-1\}$ indexes internal states.

We define two regimes:

- Under R_0: the process evolves on states $(0,j)$.
- Under R_1: the process evolves on states $(1,j)$.

In both regimes:

- The second coordinate j_t is sampled i.i.d. uniformly from $\{0, \dots, K-1\}$.
- The first coordinate i remains fixed and is never observed.

Define the observation map

$$O_m(i, j) = \mathrm{bin}_m(j),$$

the m-bit encoding of j.

The latent regime bit i is unobserved.

6.2 Equality of Observed Laws

Let $Y_t = O_m(X_t)$.

Under both R_0 and R_1,

$$Y_t \sim \mathrm{Uniform}(\{0,1\}^m)$$

independently across time.

Therefore:

$$\mathcal{L}_{R_0}(Y_{\leq b}) = \mathcal{L}_{R_1}(Y_{\leq b}) \text{ for every } b.$$

The two regimes induce identical observed processes.

6.3 Impossibility of Finite-Budget Identification

Let T_m be any test based solely on the observed prefix $Y_{\leq b}$.

Since the distributions coincide,

$$\Pr_{R_0}(\mathsf{T}_m = 1) = \Pr_{R_1}(\mathsf{T}_m = 1).$$

Under a uniform prior on $\{R_0, R_1\}$, the minimal achievable error probability is

$$\inf_{\mathsf{T}_m} \mathcal{E}(\mathsf{T}_m) = \frac{1}{2}.$$

Thus no procedure based on O_m and budget b can distinguish the regimes with nontrivial confidence.

6.4 Consequence for Structural Gap

In this construction,

$$\mathbb{E}_{R_0}[N_m(t)] = \mathbb{E}_{R_1}[N_m(t)] \ \ \forall t.$$

Hence

$$g(m) = 0.$$

This establishes:

Theorem 6.1 (Necessity of Structural Separation)

If $g(m) = 0$, then for every budget $b$and every test T_m,

$$\mathcal{E}(\mathsf{T}_m) \geq \frac{1}{2}.$$

In particular, finite-budget identifiability at scale m is impossible.

6.5 Structural Interpretation

The impossibility result is purely observational:

- The hidden dynamics differ structurally.
- The observation map erases the distinguishing component.
- No amount of computation compensates for observational collapse.

Thus identifiability is constrained not only by computational complexity, but by structural observability.

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7. Critical Scale and Structural Trade-offs

The upper and lower bounds of Sections 5 and 6 establish that identifiability at scale m depends on the interaction between latency, estimation cost, and structural separation.

We now analyze how this interaction behaves as the observational scale varies.

7.1 The Critical Identification Time

Recall the critical identification time

$$t^*(m;\delta) = t_0(m) + C\,\frac{w(m)}{g(m)^2}\log\frac{1}{\delta}.$$

This quantity represents the minimal observation horizon required for regime identification at confidence level $1-\delta$.

Scale m is identifiable under budget b if and only if:

$$t^*(m;\delta) \le b.$$

Thus identifiability reduces to comparing structural growth in m with finite resources.

7.2 Monotonicity Effects

In many observational refinement processes, the following tendencies arise:

- $t_0(m)$ is non-decreasing in m
  (finer scales delay stabilization);
- $w(m)$ is non-decreasing
  (higher resolution increases estimation cost);
- $g(m)$ is non-increasing
  (separation may dilute across dimensions).

Under these monotonicity assumptions, $t^*(m;\delta)$ is non-decreasing in m.

This implies that increasing observational resolution may eventually render identification impossible under fixed budget.

7.3 Structural Trade-off

The critical time decomposes into three competing components:

$$t^*(m;\delta) = \underbrace{t_0(m)}_{\text{latency}} + \underbrace{C\,\frac{w(m)}{g(m)^2}}_{\text{estimation/separation}}\log\frac{1}{\delta}.$$

Three structural regimes naturally emerge:

(i) Latency-Dominated Domains

If $t_0(m)$ grows rapidly with m, then:

$$t^*(m;\delta) \approx t_0(m).$$

In such domains, structural differences manifest slowly at fine scales.
Budget exhaustion occurs before separability becomes observable.

(ii) Estimation-Dominated Domains

If $w(m)$ dominates while $g(m)$ remains moderate, then

$$t^*(m;\delta) \sim \frac{w(m)}{g(m)^2}.$$

The barrier is statistical rather than structural.

Long integration windows are required even when separation exists.

(iii) Separation-Dominated Collapse

If $g(m) \to 0$ as m increases, then

$$t^*(m;\delta) \to \infty.$$

In this regime, finer observation destroys effective separability.

This produces a sharp structural boundary beyond which identification becomes impossible under any finite budget.

The phenomenon of resolution-induced collapse echoes rate–error trade-offs in communication-constrained testing: increasing resolution beyond a structural threshold may destroy finite-budget identifiability.

7.4 Existence of a Critical Scale

Define the feasible scale set:

$$\mathcal{M}(b,\delta) = \{m \in \mathbb{N} : t^*(m;\delta) \le b\}.$$

Under monotonicity of $t^*(m;\delta)$, the set $\mathcal{M}(b,\delta)$ is an initial segment of the integers.

Hence there exists a maximal identifiable scale:

$$m_c(b,\delta) = \max \mathcal{M}(b,\delta),$$

possibly zero.

This scale represents a structural boundary imposed by finite budget.

In contrast to fixed-budget best-arm identification, where allocation policies are optimized **(Degenne, 2023; Lalitha et al., 2023)**, here the allocation variable is the representation scale itself. The constraint arises before algorithmic choice.

7.5 Scale-Relative Identifiability

The analysis reveals that identifiability is not a binary property of the process.

It depends jointly on:

- observational resolution m,
- budget b,
- confidence level δ.

Increasing resolution may increase descriptive richness but simultaneously eliminate identifiability under fixed resources.

Thus finite-budget inference is inherently scale-relative.

7.6 Structural Boundary Interpretation

The existence of $m_c(b, \delta)$implies:

- For $m \leq m_c(b, \delta)$: identification is structurally achievable.
- For $m > m_c(b, \delta)$: identification is structurally impossible under budget b.

This boundary arises from observational constraints alone and is independent of computational strategy.

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8. Optimal Allocation under Finite Budget

The previous section established the existence of a maximal identifiable scale under finite budget. We now formalize the allocation problem and derive a structural optimality principle.

8.1 Problem Formulation

Fix:

- finite budget b,
- confidence level $1 - \delta$,
- structural quantities $t_0(m), w(m), g(m)$,
- and critical time

$$t^*(m;\delta) = t_0(m) + C\,\frac{w(m)}{g(m)^2}\log\frac{1}{\delta}.$$

A scale m is identifiable under budget b if

$$t^*(m;\delta) \le b.$$

Define the feasible scale set

$$\mathcal{M}(b,\delta) = \{m \in \mathbb{N} : t^*(m;\delta) \le b\}.$$

The allocation problem consists in selecting an observational scale m that maximizes structural resolution while remaining within $\mathcal{M}(b,\delta)$.

8.2 Existence of a Maximal Identifiable Scale

Under the monotonicity conditions discussed in Section 7,

- $t_0(m)$ non-decreasing,
- $w(m)$ non-decreasing,
- $g(m)$ non-increasing (or not increasing sufficiently to offset cost),

the function $t^*(m;\delta)$ is non-decreasing in m.

Proposition 8.1 (Maximal Identifiable Scale)

For every finite budget b and confidence level δ,
there exists a maximal scale

$$m_c(b,\delta) = \max\ \mathcal{M}(b,\delta),$$

possibly zero.

Interpretation

- For $m \le m_c(b,\delta)$: regime identification is structurally feasible.
- For $m > m_c(b,\delta)$: identification is impossible under budget b.

This boundary depends only on observational structure and budget, not on algorithmic design.

8.3 Structural Allocation Principle

We formalize observational resolution as utility:

$$\mathcal{U}(m) = m,$$

interpreting higher scale as higher descriptive richness.

The optimal allocation problem becomes

$$\max_{m \in \mathcal{M}(b,\delta)} m.$$

Under monotonicity, the solution is

$$m^\star = m_c(b, \delta).$$

Structural Allocation Principle

Under finite budget b and confidence level δ,
the maximal observational resolution compatible with regime identification is $m_c(b, \delta)$.

Resolution beyond $m_c(b, \delta)$ destroys identifiability, independently of computational power.

8.4 Interior Optima and Structural Classes

Depending on the relative growth of $t_0(m)$, $w(m)$, and $g(m)$, different structural classes arise.

(i) Latency-Dominated Domains

If $t_0(m)$ grows rapidly,
$m_c(b, \delta)$ saturates early.

Identification is limited by delayed structural manifestation.

(ii) Estimation-Dominated Domains

If $w(m)$ dominates while $g(m)$ remains moderate,
$m_c(b, \delta)$ scales sublinearly with b.

Identification is limited by statistical integration cost.

(iii) Separation-Dominated Collapse

If $g(m)$ decays rapidly with m,
$m_c(b, \delta)$ collapses sharply.

Finer resolution eliminates effective separability.

These structural classes are solver-agnostic and arise solely from observational geometry.

8.5 Structural Consequence

Finite-budget inference is constrained not only by computational complexity, but by structural identifiability limits induced by observational scale.

The optimal resolution under resource constraints is determined by:

- how long structure takes to reveal itself,
- how much integration is required,
- how distinguishable regimes remain at increasing scale.

These constraints are intrinsic to the pair (X_t, O_m) and persist independently of algorithmic sophistication.

Privacy-constrained identification similarly induces structural degradation of discriminability **(Chen et al., 2024)**. In our setting, representation scale plays a role analogous to privacy noise, but without stochastic perturbation-degradation arises from deterministic coarsening.

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9. Discussion

The results of this paper establish a structural theory of regime identifiability under finite observation and budget constraints. We now discuss broader implications and conceptual consequences.

9.1 Identifiability as a Structural Constraint

The main contribution is the equivalence:

$$\text{Finite-budget identifiability} \Leftrightarrow \text{Observational separability.}$$

This equivalence shows that identification limits arise not merely from computational complexity, but from structural properties of the induced observation map.

In particular:

- computational power cannot compensate for observational collapse;
- increased resolution may reduce identifiability;
- resource allocation must account for structural latency and separation decay.

These limits are intrinsic to the pair (X_t, O_m).

9.2 Beyond Complexity-Theoretic Barriers

Traditional limitations in computer science are typically framed in terms of:

- worst-case time complexity,
- hardness assumptions,
- oracle separations.

The present results highlight a distinct barrier:

structural identifiability under bounded observation.

This barrier may arise even when the underlying process is computationally simple.

Thus, identifiability and computational tractability are orthogonal notions.

9.3 Observational Resolution as a Double-Edged Sword

A key structural phenomenon is the existence of a maximal identifiable scale $m_c(b, \delta)$.

Increasing resolution:

- increases representational richness,
- but may simultaneously increase latency,
- enlarge estimation windows,
- and dilute structural separation.

Beyond the critical scale, identification becomes impossible under fixed budget.

This trade-off is structural and solver-agnostic.

9.4 Structural Classes of Domains

The classification introduced in Section 8 (latency-dominated, estimation-dominated, separation-dominated) suggests a principled taxonomy of domains based on their observational geometry.

This taxonomy is independent of algorithmic implementation and depends solely on:

- how structure manifests,

- how rapidly novelty stabilizes,
- how separability behaves under refinement.

Such classification may inform principled scale selection and diagnostic analysis in search, exploration, and adaptive inference settings.

9.5 Relation to Broader Themes

The analysis connects to several established themes:

- finite-sample hypothesis testing,
- model identifiability,
- bias–variance trade-offs,
- statistical limits of learning.

However, the present formulation differs in that it:

- operates directly on observable novelty curves,
- is scale-relative,
- explicitly incorporates latency,
- and isolates structural separation as the decisive quantity.

The lower bound construction demonstrates that observational indistinguishability can persist despite hidden structural differences.

9.6 Multi-Regime Extensions

The two-regime case analyzed here extends naturally to finite families of regimes. In such cases, identifiability depends on the minimal pairwise structural gap across regimes.

We leave a systematic multi-regime treatment to future work.

9.7 Conceptual Perspective

Finite-budget search is constrained not only by time, but by how structure becomes visible under projection.

The novelty growth curve $N_m(t)$ serves as an observable fingerprint of this interaction.

Understanding its behavior allows one to distinguish between:

- genuine structural collapse,
- statistical saturation,

- and resolution-induced indistinguishability.

This provides a structural lens for analyzing bounded observational processes.

9.8 Related Work

Finite-sample hypothesis testing under informational constraints has been extensively studied in recent years. In distributed and communication-constrained settings, fundamental limits arise when the observer receives quantized or rate-limited summaries of data **(e.g., Szabó et al., 2023; Inan et al., 2022)**. In those models, performance degradation is driven by communication bottlenecks. In contrast, the present work treats the *representation scale itself* as the structural bottleneck, even in the absence of communication constraints.

Novelty growth and support discovery have also been analyzed in unseen-species and support-size estimation problems **(Hao et al., 2020; Chakraborty et al., 2023)**. Those works focus on accurate estimation or prediction of unseen mass under i.i.d. or oracle sampling models. Here, novelty growth is not an estimation target per se, but a structural discriminator used to separate dynamical regimes under bounded observation.

In dependent settings, window-based concentration techniques have been employed to control estimation error under mixing assumptions **(Pananjady et al., 2024)**. Our integration window $w(m)$ abstracts this necessity at a structural level, without committing to a specific stochastic model.

The finite-budget identification literature in bandits **(e.g., Degenne, 2023; Lalitha et al., 2023)** studies optimal sampling allocation across arms under fixed resource constraints. In contrast, we do not optimize sampling policies; instead, we study identifiability limits induced by the choice of representation scale prior to algorithmic allocation.

Finally, recent work on privacy-constrained identification **(Chen et al., 2024)** shows that noise injection can fundamentally degrade discriminability. In our setting, degradation arises deterministically through representation coarsening, leading to a notion of maximal identifiable scale under finite budget.

Taken together, these connections situate the present work at the intersection of finite-sample testing, constrained observation, and structural identifiability. The contribution here is to isolate representation scale as a primary structural variable governing regime identifiability under finite budget.

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10. Conclusion

We introduced a non-asymptotic structural theory of regime identification under finite observational scale and budget constraints.

Our main results establish:

1. A sufficient condition for identification via a critical time bound.

2. A matching necessity result via explicit indistinguishability construction.
3. The existence of a maximal identifiable scale under finite budget.
4. A structural allocation principle for observational resolution.

These results demonstrate that identifiability is governed by latency, integration cost, and separation strength, independently of algorithmic design.

Finite-budget inference is therefore limited not only by computation, but by structural observability.

Appendix A. Proofs (Non-asymptotic)

A.1 Notation and basic objects

Fix an observational scale m and write $Y_t = O_m(X_t) \in \{0,1\}^m$.
Recall

$$N_m(t) = |\{Y_1, \dots, Y_t\}| \text{ and } \Delta_m(t) = N_m(t+1) - N_m(t) \in \{0,1\}.$$

Thus $\Delta_m(t) = 1$ iff the observation at time $t+1$ belongs to a bucket not seen in the prefix $Y_{\leq t}$.

Throughout the appendix, $R \in \{R_0, R_1\}$ denotes the regime, and $\mathbb{E}_R[\cdot]$, $\Pr_R(\cdot)$ denote expectation/probability under regime R.

A.2 A block concentration assumption

To obtain non-asymptotic guarantees without committing to i.i.d. dynamics, we use a block concentration condition.

Assumption A.1 (Block concentration for novelty increments).
There exists a window length $w(m) \in \mathbb{N}$ and constants $c_1, c_2 > 0$ such that for all $t \geq t_0(m)$ and all integers $L \geq 1$, letting

$$S_{t,L} := \sum_{s=t}^{t+L-1} \Delta_m(s),$$

we have the concentration inequality

$$\Pr_R\left(|S_{t,L} - \mathbb{E}_R[S_{t,L}]| \geq \eta\right) \leq 2\exp\left(-\frac{c_1\eta^2}{c_2\, L\, w(m)}\right) \text{for all } \eta > 0,$$

uniformly over $R \in \{R_0, R_1\}$.

Remarks.

1. Since $\Delta_m(s) \in \{0,1\}$, this assumption is compatible with many standard settings (i.i.d., mixing Markov chains, martingale-difference sequences with bounded increments), with $w(m)$ capturing the effective correlation length.
2. The role of $w(m)$ is to encode how much temporal integration is needed before empirical novelty stabilizes.

A.3 A separation hypothesis expressed at a finite time

Recall the structural gap

$$g(m) = \sup_{t \le b} \mid \mathbb{E}_{R_1}[N_m(t)] - \mathbb{E}_{R_0}[N_m(t)] \mid.$$

For the upper bound it is convenient to fix a time where the gap is realized.

Assumption A.2 (Realized separation time).
There exists $t^\dagger = t^\dagger(m) \le b$ such that

$$\mid \mathbb{E}_{R_1}[N_m(t^\dagger)] - \mathbb{E}_{R_0}[N_m(t^\dagger)] \mid \ \ge \ g(m).$$

(We may take $t^\dagger$ to be an argmax, or any time achieving at least $g(m)$.)

A.4 Proof of the finite-budget sufficiency bound (Theorem 5.1)

We restate the theorem in a form aligned with the proof.

Theorem A.3 (Finite-budget sufficiency; restatement of Theorem 5.1)

Fix scale m and confidence level $1 - \delta$. Assume A.1 and A.2.
Define the critical time

$$t^*(m; \delta) = t_0(m) + C \frac{w(m)}{g(m)^2} \log \frac{1}{\delta},$$

for a universal constant $C > 0$ depending only on the constants in Assumption A.1.
If $b \ge t^*(m; \delta)$, then there exists a test T_m such that

$$\mathcal{E}(\mathsf{T}_m) \le \delta.$$

Proof

Step 1: construct a test at time $t^\dagger$.
Let $t^\dagger \leq b$ be as in Assumption A.2. Consider the statistic $N_m(t^\dagger)$, computed from the observed prefix. Define the threshold

$$\theta = \frac{1}{2}(\mathbb{E}_{R_0}[N_m(t^\dagger)] + \mathbb{E}_{R_1}[N_m(t^\dagger)]).$$

Define the test:

$$\mathsf{T}_m(Y_{\leq b}) = \begin{cases} 1 & \text{if } N_m(t^\dagger) \geq \theta, \\ 0 & \text{otherwise.} \end{cases}$$

(If the inequality direction is reversed because $\mathbb{E}_{R_1}[N]$ is smaller, swap labels; the bound is symmetric.)

Step 2: reduce error to concentration of $N_m(t^\dagger)$.
Assume w.l.o.g. that

$$\mathbb{E}_{R_1}[N_m(t^\dagger)] - \mathbb{E}_{R_0}[N_m(t^\dagger)] \geq g(m).$$

Then:

- under R_0, an error occurs if $N_m(t^\dagger) \geq \theta$, i.e.

$$N_m(t^\dagger) - \mathbb{E}_{R_0}[N_m(t^\dagger)] \geq \frac{g(m)}{2};$$

- under R_1, an error occurs if $N_m(t^\dagger) < \theta$, i.e.

$$\mathbb{E}_{R_1}[N_m(t^\dagger)] - N_m(t^\dagger) \geq \frac{g(m)}{2}.$$

Hence, in both regimes,

$$\Pr_R(\text{error}) \leq \Pr_R\left(| N_m(t^\dagger) - \mathbb{E}_R[N_m(t^\dagger)] | \geq \frac{g(m)}{2}\right).$$

Step 3: express $N_m(t^\dagger)$ as a sum of increments after $t_0(m)$.
Write

$$N_m(t^\dagger) = N_m(t_0) + \sum_{s=t_0}^{t^\dagger - 1} \Delta_m(s), t_0 := t_0(m).$$

The term $N_m(t_0)$ is measurable at time t_0 and contributes no randomness beyond that prefix; concentration is thus controlled by the sum of increments. Let

$$L := t^\dagger - t_0.$$

Then

$$N_m(t^\dagger) - \mathbb{E}_R[N_m(t^\dagger)] = (\sum_{s=t_0}^{t^\dagger-1} \Delta_m(s)) - \mathbb{E}_R[\sum_{s=t_0}^{t^\dagger-1} \Delta_m(s)].$$

Step 4: apply block concentration (Assumption A.1).
Assumption A.1 yields for $\eta = g(m)/2$:

$$\Pr_R(\text{error}) \le 2\exp\left(-c_1 \frac{(g(m)/2)^2}{c_2\, L\, w(m)}\right) = 2\exp\left(-\frac{c_1}{4c_2} \frac{g(m)^2}{L\, w(m)}\right).$$

We want this to be $\le \delta$. It suffices that

$$\frac{c_1}{4c_2} \frac{g(m)^2}{L\, w(m)} \ \ge\ \log \frac{2}{\delta}.$$

Equivalently,

$$L \ \ge\ \underbrace{\frac{4c_2}{c_1}}_{=:C} \frac{w(m)}{g(m)^2} \log \frac{2}{\delta}.$$

Step 5: conclude via budget constraint.
If

$$b \ge t_0(m) + C \frac{w(m)}{g(m)^2} \log \frac{2}{\delta},$$

then choosing $t^\dagger \le b$ (or simply $t^\dagger = b$ if separation is realized by b) yields $\mathcal{E}(\mathsf{T}_m) \le \delta$.
Absorbing constants into C and $\log\left(\frac{2}{\delta}\right) \asymp \log\left(\frac{1}{\delta}\right)$ gives the stated form.

□

A.5 Proof of the maximal identifiable scale (Proposition 8.1)

We restate the monotonicity-based claim.

Proposition A.4 (Existence of maximal identifiable scale; restatement of Proposition 8.1)

Fix $b \in \mathbb{N}$ and $\delta \in (0,1)$. Suppose $t^*(m;\delta)$ is non-decreasing in m. Define

$$\mathcal{M}(b,\delta) = \{m \in \mathbb{N} : t^*(m;\delta) \leq b\}.$$

Then $\mathcal{M}(b,\delta)$ is either empty or of the form $\{1,2,\dots,m_c\}$ for some $m_c \in \mathbb{N}$. In particular, there exists a maximal identifiable scale

$$m_c(b,\delta) = \max\ \mathcal{M}(b,\delta),$$

possibly zero (if the set is empty).

Proof

If $\mathcal{M}(b,\delta) = \emptyset$, set $m_c(b,\delta) = 0$. Otherwise pick any $m \in \mathcal{M}(b,\delta)$. Since $t^*(\cdot;\delta)$ is non-decreasing, for every $m' \leq m$ we have

$$t^*(m';\delta) \leq t^*(m;\delta) \leq b,$$

so $m' \in \mathcal{M}(b,\delta)$. Hence $\mathcal{M}(b,\delta)$ is an initial segment of $\mathbb{N}$, and its maximum exists because it is bounded above by any m such that $t^*(m;\delta) > b$.

□

A.6 Lower Bound via Observational Indistinguishability

Lemma A.5 (Lower bound under identical observed laws).
Let R_0, R_1 be two structural regimes and fix representation O_k and budget b.
Denote by

$$\mathcal{L}_i^{(k,b)}$$

the induced law of the observed prefix

$$\mathbf{Y}_{\leq b}^{(k)} = (Y_1,\dots,Y_b), Y_t = O_k(X_t),$$

under regime R_i, for $i \in \{0,1\}$.

If

$$\mathcal{L}_0^{(k,b)} = \mathcal{L}_1^{(k,b)},$$

then for every (possibly randomized) decision rule

$$\phi: \mathcal{Y}^b \to \{0,1\},$$

the minimax error satisfies

$$\inf_{\phi} \max \{\mathbb{P}_{R_0}^{(k,b)}(\phi(\mathbf{Y}_{\leq b}^{(k)}) = 1), \mathbb{P}_{R_1}^{(k,b)}(\phi(\mathbf{Y}_{\leq b}^{(k)}) = 0)\} \geq \frac{1}{2}.$$

Proof

If the induced laws coincide, then for any measurable test ϕ,

$$\mathbb{P}_{R_0}^{(k,b)}(\phi = 1) = \mathbb{P}_{R_1}^{(k,b)}(\phi = 1).$$

Hence,

$$\mathbb{P}_{R_0}^{(k,b)}(\phi = 1) + \mathbb{P}_{R_1}^{(k,b)}(\phi = 0) = \mathbb{P}_{R_0}^{(k,b)}(\phi = 1) + (1 - \mathbb{P}_{R_1}^{(k,b)}(\phi = 1)) = 1.$$

Therefore at least one of the two error probabilities is $\geq \frac{1}{2}$, and taking the infimum over ϕ yields the claim.

□

Structural interpretation

If two regimes are observationally indistinguishable under (O_k, b), no test—regardless of computational power—can outperform random guessing.

Appendix B. Structural Examples and Regime Geometry

This appendix provides explicit processes illustrating the structural quantities:

- latency $t_0(m)$,
- window length $w(m)$,
- separation gap $g(m)$,
- critical time $t^*(m; \delta)$.

All examples are solver-agnostic and operate directly on observed sequences $Y_t = O_m(X_t) \in \{0,1\}^m$.

B.1 A Stabilizing (Core-like) Regime

We construct a regime where novelty collapses after a finite transient.

Definition

Fix scale m. Let:

- For $t \leq t_0(m)$: Y_t are distinct.
- For $t > t_0(m)$: $Y_t = Y_{t_0(m)}$ deterministically.

Then:

$$N_m(t) = \begin{cases} t & t \leq t_0(m), \\ t_0(m) & t > t_0(m). \end{cases}$$

Structural quantities

- Latency: $t_0(m)$ explicit.
- Window: $w(m) = O(1)$ (no stochastic integration needed).
- Separation: depends entirely on difference in transient length.

If two regimes differ only in $t_0(m)$, then

$$g(m) = \mid t_0^{(1)}(m) - t_0^{(0)}(m) \mid.$$

Interpretation

This is a strict collapse regime.
Beyond $t_0(m)$, no new information is generated.

Finite-budget identification is possible only if b exceeds the larger latency.

B.2 A Linearly Expanding Regime

We now consider a process with steady novelty growth.

Definition

At each step, with probability $p(m) \in (0,1)$,
a new unseen bucket is generated; otherwise, a previously seen bucket is repeated uniformly.

Then:

$$\mathbb{E}[N_m(t)] \approx 1 + p(m)\, t.$$

Structural quantities

- $t_0(m) = O(1)$.
- $w(m) = O\left(\frac{1}{p(m)}\right)$ (law of large numbers window).
- If regimes differ in $p(m)$, then

$$g(m) \asymp \mid p_1(m) - p_0(m) \mid \; t.$$

Consequence

Separation grows linearly in time.

Identification is limited primarily by statistical integration:

$$t^*(m;\delta) \asymp \frac{w(m)}{g(m)^2} \log \frac{1}{\delta}.$$

If $p(m)$decreases with m, separation may shrink under refinement.

B.3 Exponential Saturation (Soft Collapse)

Consider:

$$\mathbb{E}[N_m(t)] = M(m)(1 - e^{-\lambda(m)t}),$$

with $M(m)$ maximal distinct buckets.

Structural behavior

- Early exponential growth,
- Saturation to $M(m)$.

If two regimes differ only slightly in $\lambda(m)$, then:

$$g(m) \approx \frac{M(m)}{e} \; \mid \lambda_1(m) - \lambda_0(m) \mid \; t^{\dagger}.$$

If $\lambda(m) \to 0$ as $m \to \infty$, separation collapses.

Interpretation

This models domains with finite but large representational capacity.

High resolution increases $M(m)$, but slows convergence.

B.4 Separation-Dominated Collapse

We now build an example where refinement destroys identifiability.

Construction

Let there be two hidden regimes that differ only in a coordinate $j(m)$ that appears with probability:

$$q(m) = 2^{-m}.$$

Thus:

- At coarse scale m, the difference is observable.
- As m increases, the distinguishing event becomes exponentially rare.

Then:

$$g(m) \asymp q(m) = 2^{-m}.$$

Hence:

$$t^*(m;\delta) \asymp \frac{w(m)}{2^{-2m}} \log \frac{1}{\delta} = w(m)\, 4^m \log \frac{1}{\delta}.$$

Finite budgets impose a sharp maximal identifiable scale:

$$m_c(b,\delta) \approx \frac{1}{2} \log_2 \left(\frac{b}{w(m)} \right).$$

Interpretation

Refinement reveals more coordinates but dilutes discriminative signal.
This is structural collapse due to resolution.

B.5 Non-Monotonic Identifiability

We now show that monotonicity of $t^*(m)$ is not guaranteed.

Example

Let:

- For small m: separation weak.
- For intermediate m: separation increases (hidden feature revealed).
- For large m: separation decays (rare event dilution).

Then $g(m)$ may be unimodal.

If $w(m)$ grows slowly,
$t^*(m)$ can exhibit interior minima.

Consequence

An interior optimal scale may exist:

$$m^\star \neq 1, m^\star \neq m_{\max}.$$

This justifies Section 8.4 beyond purely monotone assumptions.

B.6 Chaotic Alternation

Consider alternating structural regimes:

- blocks of collapse,
- blocks of expansion.

Let:

$$\mathbb{E}[N_m(t)] = \begin{cases} \text{linear growth} & t \in I_1, \\ \text{flat} & t \in I_2, \\ \text{linear} & t \in I_3. \end{cases}$$

Finite windows may misclassify.

This shows:

- local stabilization does not imply global collapse,
- window size $w(m)$ is critical.

B.7 Structural Takeaways

Across examples:

- $t_0(m)$ encodes manifestation latency.
- $w(m)$ encodes effective integration cost.
- $g(m)$ encodes discriminative geometry.

Different domains may be:

- latency-dominated,
- estimation-dominated,

- separation-dominated,
- or mixed.

Finite-budget identifiability is a geometric property of this triple.

Appendix C. Artifact and Empirical Structural Estimation

This appendix describes the reference implementation accompanying the paper. The artifact operationalizes the structural quantities introduced in Sections 2–8 and provides empirical estimates of identifiability under finite budget.

The code is illustrative and solver-agnostic: it operates on observed sequences $Y_t \in \{0,1\}^m$.

C.1 Observable Inputs

The artifact assumes access to:

- A sequence of observed vectors

$$Y_1, Y_2, \dots, Y_b \in \{0,1\}^m$$

obtained from an external process.

- A chosen observational scale m.

No assumptions are made on the generator of Y_t.

C.2 Empirical Novelty Curve $\widehat{N}_m(t)$

The artifact computes:

$$\widehat{N}_m(t) = |\{Y_1, \dots, Y_t\}|.$$

This is the empirical counterpart of the theoretical novelty curve $N_m(t)$.

The incremental process:

$$\widehat{\Delta}_m(t) = \widehat{N}_m(t+1) - \widehat{N}_m(t) \in \{0,1\}$$

is also recorded.

C.3 Empirical Window Length $\widehat{w}(m)$

To estimate the effective integration window, the artifact computes block-variance statistics.

Let

$$S_{t,L} = \sum_{s=t}^{t+L-1} \widehat{\Delta}_m(s).$$

For increasing window lengths L, the empirical variance:

$$\widehat{\mathrm{Var}}(S_{t,L})$$

is compared against linear scaling.

The minimal L for which variance growth stabilizes is taken as an estimate of $w(m)$.

Interpretation:

- small $\widehat{w}(m) \rightarrow$ near-independent increments,
- large $\widehat{w}(m) \rightarrow$ long correlation length.

C.4 Empirical Separation Gap $\widehat{g}(m)$

Given two observed trajectories (e.g., from two candidate regimes), the artifact computes:

$$\hat{g}(m) = \max_{t \le b} \mid \widehat{N}_m^{(1)}(t) - \widehat{N}_m^{(0)}(t) \mid.$$

If only one trajectory is available, bootstrapped sub-sampling provides proxy separation estimates.

This corresponds directly to the theoretical $g(m)$.

C.5 Empirical Critical Time $\hat{t}^*(m;\delta)$

Using empirical estimates:

$$\hat{t}^*(m;\delta) = \hat{t}_0(m) + C\,\frac{\widehat{w}(m)}{\hat{g}(m)^2}\log\frac{1}{\delta}.$$

Where:

- $\hat{t}_0(m)$ is estimated as the earliest time after which novelty growth becomes approximately stationary,
- C is a universal constant (set conservatively).

This provides an empirical identifiability threshold.

C.6 Identifiability Map

For each scale m, the artifact determines:

$$\widehat{m}_c(b, \delta) = \max\{m: \hat{t}^*(m; \delta) \leq b\}.$$

This yields a practical approximation of the maximal identifiable scale.

C.7 Relation to Theory

The artifact directly instantiates the structural quantities:

Theoretical Quantity	Empirical Proxy
$N_m(t)$	$\widehat{N}_m(t)$
$w(m)$	variance stabilization window
$g(m)$	maximal novelty gap
$t^*(m; \delta)$	empirical critical time

The implementation does not assume specific stochastic models. It operates purely on observed novelty behavior.

C.8 Scope and Limitations

The artifact:

- does not prove identifiability,
- does not guarantee correctness under arbitrary processes,
- assumes sufficient sample size for estimation.

It serves as a diagnostic tool illustrating structural behavior under finite budget.

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